Learning from Data – Final Project

General remarks

For the final project, we will look at the DiCo-NLI task that we have been working with. Differently than in the other assignments, we ask you to be a bit creative: you should come up with your own research question. Grading will be determined by your final report. See the final page for the specific grading sheet. **This assignment is to be done individually.**

Deadline for submission on Brightspace: Monday November 30th, 2026, 23:59.

What you have to hand in by the deadline:

- The final research report. See Section 3 for details. Please, make sure to hand in a **pdf** file following the usual report template.

1 Data

Please use the same data set as you have been using throughout the course. This means that you also use the train/dev/test split as you have so far. In the paper, make sure to still describe this data set, and the papers that the data set comes from (or was based on).

2 Research goal of the project

I want you to address two different goals in this research project.

- Creating the best possible model given what you learned so far
- Answering an additional original and creative research question

For the first goal, you are going to fine-tune language models or prompt LLMs. Do not forget to also include baseline scores of simple machine learning models. The second one is more interesting. You are expected to come up with an approach or research question that does a bit more than simply fine-tuning a language model to get a high score. You are free to choose your own research question. This can be a very small idea, and we do not expect groundbreaking research, but try to go a bit further than just reporting scores of models.

Of course, whatever question you choose, you should check if there is any previous work that did something similar.

3 What you have to do

This section contains what you have to do for the project. Please read it carefully!

Models Independent of your research questions, we ask you to **at least** implement and provide scores:

- A baseline classic model using n-grams (e.g. Naive Bayes, SVM)
- A number of fine-tuned pretrained language models (i.e. BERT, RoBERTa, DeBERTa)

Luckily you more or less did all of this already in the assignments! Make sure to provide a description of these models in your report. Of course, you are free to use any other model you want! Please experiment!

Report You are also asked to produce a **report**. Again, you have to write this report as a **research paper.** The report should start with clearly explaining the problem you are working on, motivating why this is an important problem to work on. The research questions should be clearly laid out and you should cite previous work.

The method should contain the explanation of how you tackled this problem, a description of the algorithm(s) you chose to train your model, including parameter tuning and settings, any additional data/resources you incorporated, and how well you do when developing. You should also justify your choices explaining why you selected a certain approach, certain features, the learning algorithm, and so on.

In the results section you should clearly present all relevant results. How well do your models do? Do your models perform better than expected, and if so, why? You should interpret the results if possible: how well are we actually doing on this task? Also include a section in which you discuss the results, with possibly an extra (error) analysis. End the paper with a clear conclusion and possible suggestions for future work.

Code There is no need to submit your code for the final project. However, you could potentially add a link to a public GitHub repository that contains the code for the project. This is good practice, anyway.

Grading sheet See below for the full grading sheet. You will get a grade for each criterion: the final grade is the average of all these grades. So make sure to check it out!

Writing	ATROCIOUS–BAD (1–4)	INSUFFICIENT (5)	SUFFICIENT (6)	GOOD (7–8)	EXCELLENT (9–10)
Structure	lacks a clear structure	inconsistent and/or not in agreement with conventions	logically structured in conventional sections	completely in line with contents and follows conventions where necessary	excellent (suitable for publication in a scholarly journal)
Coherence & Language	incoherent and difficult to follow, many grammatical errors or typos	parts of the report are not consistently linked at the level of chapters, sections, and paragraphs, report is hard to follow, English should be improved	parts of the report are sufficiently linked at the level of chapters, sections, and paragraphs	strongly cohesive at all levels (chapters, sections, paragraphs), level of English is good	extremely cohesive at all levels; professional argumentation throughout, excellent language use
Introduction	ATROCIOUS–BAD (1–4)	INSUFFICIENT (5)	SUFFICIENT (6)	GOOD (7–8)	EXCELLENT (9–10)
Motivation & Previous work	the rationale is missing and the study is not contextualised	the rationale is unclear and contextualisation is weak	the rationale is explained sufficiently clearly and the contextualisation is sufficient	the rationale is convincing and clear, the study is well contextualised	the rationale clearly shows that the study is important and the study is very well contextualised
Research questions	there are no research questions	the research questions are unclear, too general, or not linked to topic	sufficiently clear and feasible research questions linked to topic	clear and interesting research question that follows from the data set	very clear and interesting and challenging research question that follows logically from the data set
Method	ATROCIOUS–BAD (1–4)	INSUFFICIENT (5)	SUFFICIENT (6)	GOOD (7–8)	EXCELLENT (9–10)
Description	severely lacking description of used data sets and algorithm	barely any description of used data sets and algorithms	sufficient description of data sets and algorithms but certain parts are left out	good description of data sets and algorithms used	excellent description of data sets and algorithms
Reproducbility	method is lacking import details, reproduction impossible	method is lacking certain details, which makes reproduction hard	method contains most details but can be improved	method contains all details for reproducibility but some parts are not clear or irrelevant	excellent description of the method containing all relevant details
Models	Clearly did not meet minimum requirements in terms of different models	Not enough models have been implemented, baselines are lacking	Sufficient implementation of a number of algorithms, baselines are OK	Implementation of a number of different algorithms with different feature sets or architectures, good use of baselines	Excellent use of different algorithms and feature sets/architectures, use of correct and fair baselines
Results	ATROCIOUS–BAD (1–4)	INSUFFICIENT (5)	SUFFICIENT (6)	GOOD (7–8)	EXCELLENT (9–10)
Overview	no clear overview of all results	weak (not all relevant results are shown or discussed)	sufficient overview of results, some parts still unclear or missing	good overview of all relevant results	excellent overview of results, showing all results in a logical manner
Performance	bad results (below baseline) and no explanation provided	weak results, no explanation provided	bad results with partial explanation provided, or decent results without explanation	good results with good explanation provided	excellent results, models are pushed to the best performance
Discussion & Analysis	lacking	weak (no interpretation of results)	sufficient discussion or interpretation of results, some extra analysis is performed	critical discussion of results, which are interpreted to give insights. Some nice extra analysis is performed	insightful, creative and critical discussion of results, great extra (error) analysis performed